

GenSec Industries: Project Overview document



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1. Introduction

The focus of this document is to have a central space where the entirety of the project can be found. This document will be used to refer to another document when applicable. Topics such as the context, description goal etc. of the project can all be found in this document. The aim of this document is to show a problem description and some research questions on how the problems are planned to be solved.

2. Project context

The airport crowd management system is a system that took inspiration from the occupancy management system for the industry project. Unlike the occupancy management system, the crowd management system does not manage flows within buildings to keep track and manage how many people are inside. Rather, it is to manage the flow how passengers pass through security at an airport.

The goal of the project is to guide passengers upon ticket scan from their spot to the best gate queue. Take figure 1. In this representation of queue line at an airport, where 1 blue star represents 20 people. The customer may not know that the line of gate 3 has the least number of people.

The system aims to guide customers to the line that is the least busy upon ticket scan. In this case, if when a customer scans their ticket, the system will compute and show the customer to gate 3.

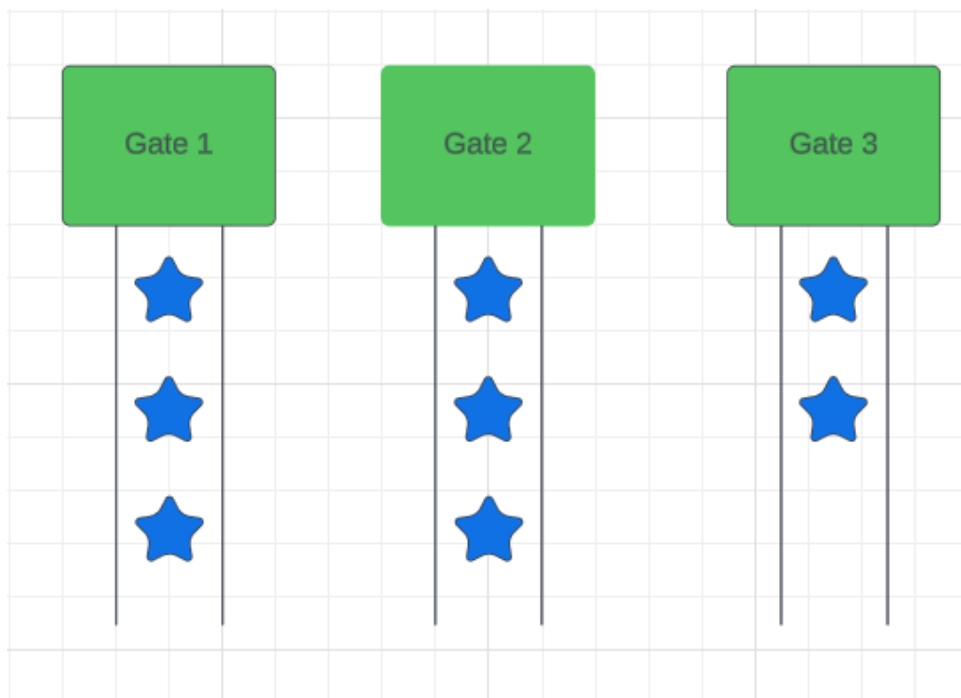


Figure 1: Representation of 3 queue lines.

3. Project description

The system to be built needs to be a decentralized system that must handle errors on its own, correct mistakes and handle the logic behind operations without the need for a central node. See figure 2 for the context diagram. This context gives a more in-depth view of how the system functions. Here a customer can scan their ticket.

The gates that are present are divided into groups. Each group has a sort of leader which is called the divider. The gates themselves do not communicate with each other but with their personal divider.

The dividers communicate with each other. Among the dividers, there will be one leader that will be performing the logic of finding the gate with the lowest traffic. In other words, the leader will perform the logic to tell the passenger, that they need to go to gate 3 in figure 1 by showing this on the UI screen.

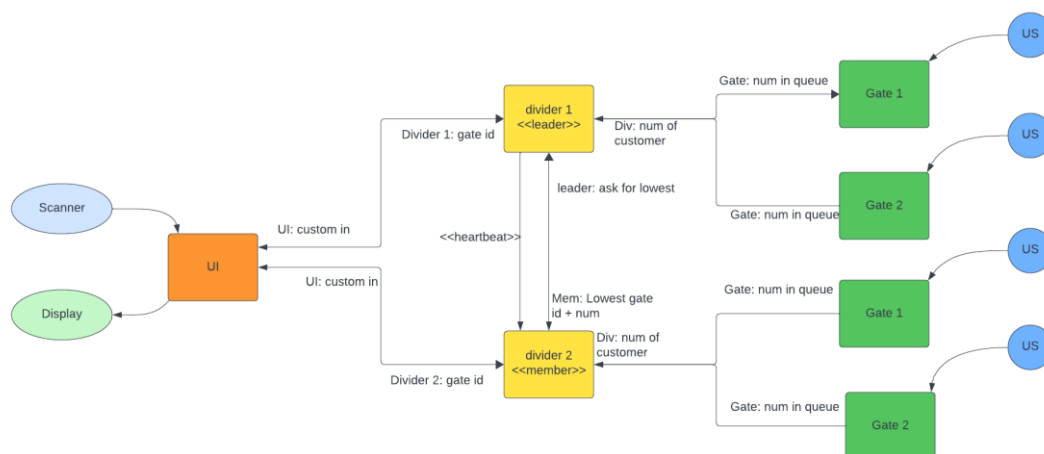


Figure 2: Context diagram

4. Project goals

The goal of this project is to solve a similar problem to what was mentioned in the occupancy management system project. The idea is to automate the flow of which passengers go to their designated gates. Instead of choosing a gate line they will be advised to go to a different gate.

This is to ensure that there aren't any unusually long wait times in the queues. It can also solve some bottlenecks seeing as the system divides the customers evenly among gate lines.

5. Methodology

The methodologies that were used to get the desired results were all taken from the DOT framework that Fontys university of Applied science has provided us with.

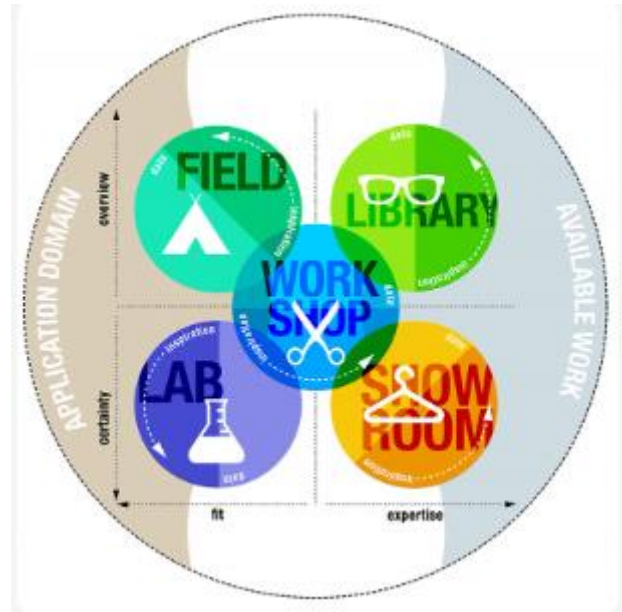


Figure 3: Dot framework.

Most of the methods in the Dot framework methods were used to conduct proper research for the system. Under these methods include:

- **LAB:** The lab methodology was used to create different proof of concepts (PoC) for the system that will later be implemented into one whole system.
- **Showroom:** The project is made by working agile. The project is then split into sprints were the developers showcase their progress, plans and concerns every set period.
- **Workshop:** The workshop methodology is used to prototype/test the project to get more insight on how the project can be improved upon.
- **Library:** The library methodology is used to gain knowledge on each topic that will be either used or prototyped within the project. The research can be catalogued so that already can be accessed for future purposes.

6. Results

The entire team was happy with the results of the project after all the hardships of changing the original design. There could be improvements in the system such as closing the gate automatically when its parent divider is dead. Adding the emergency feature could also be an improvement. The overall results were good as most of the requirements were met.

What went really went was the lead divider logic seeming as it was a device that had the responsibility of many tasks. Regarding the communication within the system, it can be classified as heavy however it is needed as big amount of information is precisely shared between devices.

What could be added extra is for example additional information for the user. For example, in the UI or the documentation etc.

7. Evaluation

It could be concluded that the team was proud of the project that was made. Most of the set goals were met. Additionally, some features that were not in the original idea was also implemented, like

the scanner. In the end, everyone is happy with what has been made and it feels like a good project that can later be improved. For example, add a user-friendly interface.

One aspect of the project that could've been done better was earlier beginning with integration and testing of the system.

8. Bibliography

[1] - Vogel, J. (n.d.). *The DOT Framework*. ICT Research Methods. <https://ictresearchmethods.nl/dot-framework/>